

04 Rate Laws

Why are we learning this?

Chemical reactor design is the process of designing reactors around **chemical kinetics**. In previous lessons, we developed mole balances and introduced conversion. However, those equations are not enough to solve reactor design problems. To predict reactor performance, we must also know how the reaction depends on the concentrations of the various reactant and product species. That information comes from the **rate law**.

By the end of today's lesson, you should understand:

- What a rate law is.
- What reaction order means.
- How to write rate laws for elementary reactions.
- How to write rate laws for nth-order reactions.
- How reversible reactions affect rate laws.
- How stoichiometry relates the rates of different species.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



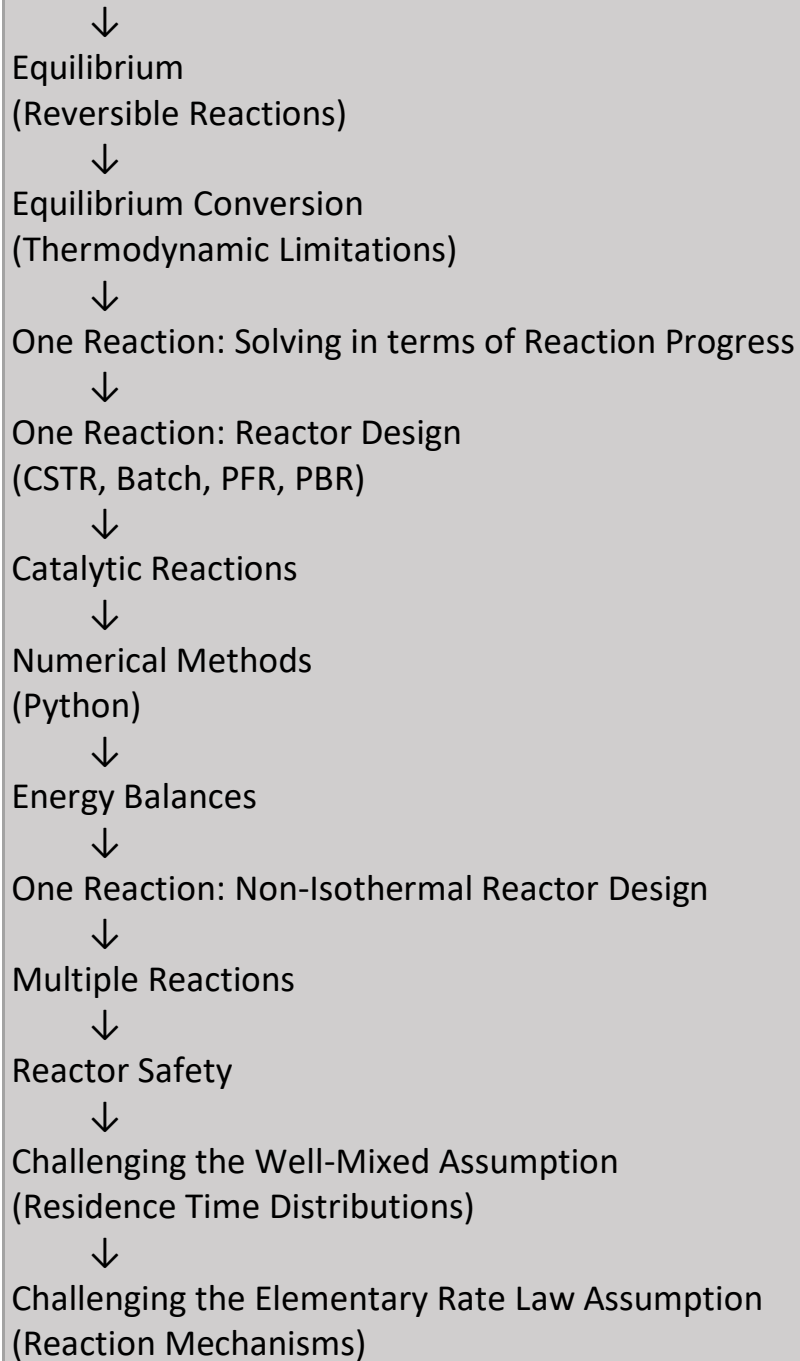
Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



► Rate Laws ◀
(Kinetics)



Big Idea

A mole balance tells us **what must be accounted for**.

A rate law tells us **how the reaction depends on the concentrations of different species**.

Both are needed to solve reactor design problems.

A rate law is

They are used to model how the net rate of generation of a species changes due to changes in

What Does a Rate Law Tell Us?

A rate law connects the reaction rate to measurable quantities, i.e., concentration or pressure.

In many systems:

- Higher reactant concentration leads to a faster reaction.
- Lower reactant concentration leads to a slower reaction.

The exact relationship depends on the reaction kinetics.

Often, the changes in concentration for a chemical reaction can be modeled using

For example, take the reaction $A + B \rightarrow C + D$. Assuming this reaction is irreversible, a good way to model the concentration dependence of the rate law is

where n & m are

Theoretically, n & m could span from

They don't have to be

Reaction Order vs Stoichiometry

Students frequently confuse these concepts.

Stoichiometric coefficients come from the balanced chemical equation.

Reaction orders describe how strongly the reaction rate depends on concentration.

Reaction orders are kinetic quantities and do **not** necessarily equal stoichiometric coefficients.

Example:



The coefficients are 2 and 1, but the rate law may or may not contain those same exponents.

In this class, we are often concerned with two specific cases:

- First specific case: The reaction is elementary.

This means that the rate law is $-r_A =$

Therefore, for the irreversible reaction $A + B \rightarrow C + D$, $-r_A =$

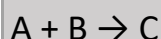
Let's do another example. The rate law for the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ is elementary. What is the rate expression?

What Does "Elementary" Mean?

An elementary reaction occurs in a single molecular event. For elementary reactions:

Reaction orders = Stoichiometric coefficients

Example:



The rate law is: $-r_{\text{A}} = k_{\text{A}} C_{\text{A}} C_{\text{B}}$

This reaction has an elementary rate law.

Two important notes pertaining to elementary reactions:

- One: If you are told that a reaction is elementary, take that to mean that the reaction is *elementary as written*. Do not alter the stoichiometry for the reaction and then write the rate expression based off the altered stoichiometry. For example, it would be incorrect to alter the stoichiometry of the reaction above to $\text{NO} + 1/2 \text{O}_2 \rightarrow \text{NO}_2$ and then write the rate law as $-r_{\text{NO}} = k_{\text{NO}} C_{\text{NO}} C_{\text{O}_2}^{1/2}$. Note that this gives entirely different concentration dependence of the rate than the expression above. If you are given a reaction and told it's elementary, it means that it's elementary as written.

- Two: If you are working homework/exam problems in this class and you aren't given information about the rate law, you can assume it's elementary. In this case, write: "assume the rate law is elementary" on your solution and proceed with the solution using the elementary rate expression.

Why "As Written" Matters

When a problem says a reaction is elementary, it is elementary **as written**.

Do not simplify or divide the stoichiometry before writing the rate law.

Changing the stoichiometry changes the implied molecular event and therefore changes the kinetics.

- Second specific case: The reaction is "Nth order".

Sometimes, you'll be told: "The reaction is N order in A" or simply, "The reaction is N order." In this case:

Often, when the kinetic information is given in this way, it means the rate expression only depends on the concentration of one component. For example:

- $A + B \rightarrow C$. The reaction is 1st order in A means

- What about this reaction: $\text{CH}_2\text{OCH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{OHCH}_2\text{OH}$. The reaction is 1st order. What is the rate law? Why?
- One more example. The reaction $2\text{A} + \text{B} \rightarrow \text{C}$ is 1st order in B and -2nd order in A. What is the rate law?

What Does Nth Order Mean?

Sometimes experiments reveal the concentration dependence of a reaction even though the underlying mechanism is unknown.

Examples:

Rate Law	Order
k	Zero order
$k\text{CA}$	First order
$k\text{CA}^2$	Second order
$k\text{CA}^{-1}$	Negative first order

Reaction orders:

- Can be negative.
- Can be fractional.
- Do not need to be integers.

- Sometimes, the rate doesn't follow a power law model. For example, the rate law for Fogler problem 5-14_B for the gas phase dehydration of butanol over a silica-alumina catalyst is $-r_A = \frac{kP_A}{(1+K_{C,A}P_A)^2}$, where P_A is the partial pressure of A (A = butanol), and $K_{C,A}$ is a constant (specifically, it is the adsorption equilibrium constant for butanol on the silica-alumina catalyst). P_A is related to C_A through the ideal gas law: $C_A = \frac{P_A}{RT}$. In this case, the rate law that you'd use in your mole balance equation is $-r_A = \frac{kC_A RT}{(1+K_{C,A}C_A RT)^2} = \frac{k' C_A}{(1+K_{P,A}C_A)^2}$ (where the relationship between K_P and K_C is given in Equations 61 and 64 in the Reference Book).
- Some tips:
 - Don't overthink construction of the rate law. Just do what the problem is telling you.
 - Always write the rate law that you're working with as part of your problem solution. That way, if you do misinterpret what the rate law should be, it can be easily seen.

Not Every Rate Law Is a Power Law

Power-law kinetics are useful because they are simple. Real reactions often involve:

- multiple reaction steps,
- Catalysts,
- transport limitations.

As a result, real rate laws can be more complicated than:

$$-r_A = kC_A^n C_B^m$$

The important skill is identifying the rate law that applies to the problem.

Let's look at another example. The reaction $A + 2B \rightleftharpoons 3C$ is reversible and exhibits elementary kinetics. What is the rate law?

Key definition: $K_C = \frac{k_f}{k_r}$ where K_C is the

Hence,

Reversible vs Irreversible Reactions

For an irreversible reaction:

Forward reaction matters.

Reverse reaction is negligible.

For a reversible reaction:

Forward reaction matters.

Reverse reaction matters.

The observed rate is the difference between the forward and reverse rates.

Kinetics Meets Thermodynamics

One of the most important relationships in reaction engineering is:

$$K_C = k_f / k_r$$

The equilibrium constant is directly related to the kinetic parameters.

This relationship forms the bridge between:

- kinetics (how fast reactions proceed), and
- thermodynamics (how far reactions can proceed).

Finally, we need to discuss relative rates. The rate expressions for different species in a chemical reaction are related to each other via stoichiometry. For example:

- For the reaction $2A + B \rightarrow C$, we said $-r_A = k_A \frac{C_B}{C_A^2}$. What are the units on $-r_A$?
- What is the relationship between $-r_A$ and $-r_B$?
- Key equation: Equation 3-1 in Fogler and Equation 45 in the Reference Book: For the reaction $aA + bB \rightarrow cC + dD$:

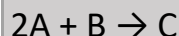
$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d}$$
- Using this equation, what is r_C for the elementary, reversible reaction $A + 2B \rightleftharpoons 3C$, given $-r_A = k_{A,f} C_A C_B^2 - k_{A,r} C_C^3$?

One Reaction, Multiple Species

A chemical reaction consumes and produces several species simultaneously.

Because reactions obey stoichiometry, all species rates are related.

For:



every:

- 2 mol A consumed
- 1 mol B consumed
- 1 mol C produced

occur together.

Knowing one species rate allows all other rates to be determined.

Connecting Rate Constants and Reactor Design

Recall that Levenspiel plots use:

$$\frac{F_{A0}}{-r_A}$$

If:

$$-r_A = kC_A^n$$

then increasing conversion decreases $-r_A$ and the Levenspiel plot has a positive slope. A PFR will be more efficient than a CSTR.

But what if $-r_A = \frac{k}{C_A^n}$?

Then, increasing conversion *increases* $-r_A$. The Levenspiel plot has a *negative* slope. In this case, a CSTR will be more efficient than a PFR.

Summary and Key Takeaways

Today we introduced rate laws and their role in reactor design.

Key Ideas

- ✓ Mole balances are only half of a reactor design problem. A rate law is also required.
- ✓ Elementary reactions are special. For elementary reactions: Reaction orders = stoichiometric coefficients.
- ✓ Reaction orders are kinetic quantities. They do not necessarily equal stoichiometric coefficients.
- ✓ Not all reactions follow simple power-law expressions. More complex kinetic models are common in practice.
- ✓ Rates of all species within a single chemical reaction are linked by stoichiometry. Knowing one species rate allows all others to be determined.

Most Important Idea

Mole Balance

+

Rate Law

=

Reactor Design Equation

Understanding how to construct and interpret rate laws is a foundational skill in chemical reaction engineering.